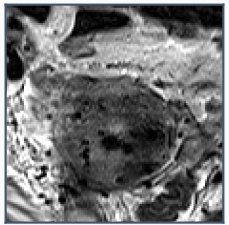
NeedleFinder 2026: Sub-Millimeter 3D Catheter Reconstruction on 3T MRI for Interstitial Brachytherapy in 4.15 s

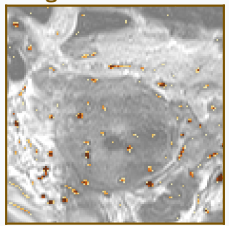
1. Clinical Bottleneck (45–50 min manual)

15–30 6F (2 mm) plastic tubes = dark voids on 3T MRI

Native 3T T2W MRI



Stage 0 Dark-Core S0



- **54 BWH patients (1,063 tubes, Δz = 1.2–6.0 mm)**
- 23-pt severe adhesion (≤3 mm) · 40 dual-rater
- 38 paired CT · 48 zero-needle · 243 multi-seq

2. Physics-Guided Graph & Conformal Pipeline (4.15 s CPU)

Stage 0: 3D Dark-Core Symmetry + Auto-Crop (0.27 s CPU)

6 axes × 4 radii (1.35–3.3 mm) + k_lim_auto (48× fewer FPs)

Stage 1: Native 3D ResUNet + 51-Scan SSL (2.58 s CPU)

Sub-pixel peaks P_peak resolve 2.2 mm twins + obturator P_obt

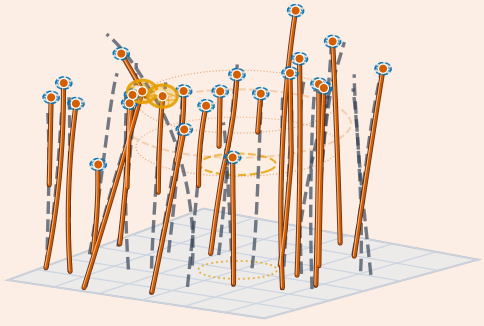
Stage 2A/2B: Triplet MPNN + Stitch + Bundle GNN (1.25 s)

4-layer bending GNN + ≤20 mm stitcher + SynthBrachy3T GNN

Stage 2C: 1D Tip Localizer + Conformal Spheres (0.06 s)

91-step (0.5 mm) 1D ResNet + 90% 3D sphere (r90 = q90·σ)

3. 3D Bundle, Conformal Tips & TG-43 Dose



- **91.4% Recall · 89.4% F1 (+6.3 to +50.7 vs 6 methods)**
- **0.47 mm shaft (0.44 mm long) · 2.26 mm 3D tip**
- Non-inferior to dual experts; 52% FPs = real on CT
- **18 s* 2-tip check → 2.8% |ΔD90| · Prostate 96.3% F1**